

Imaging in the Presence of Magnetic Field Inhomogeneities

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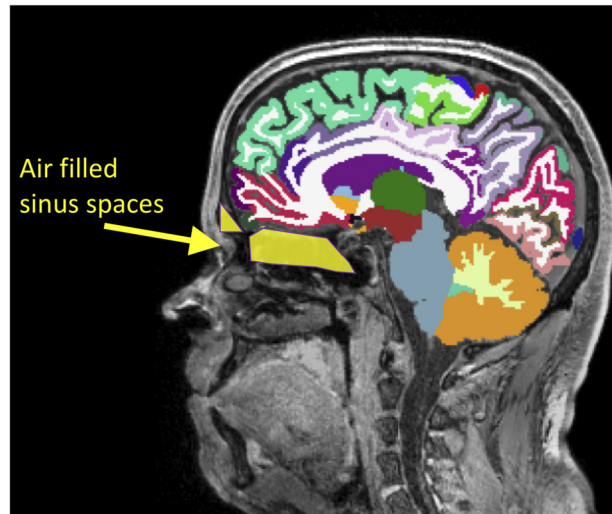
12.1 Introduction

As seen in Chapter 1, accurate spatial localization of spins with MRI relies on a known relationship between the location of a spin in the three-dimensional (3D) space and the magnetic field strength at that location. In the case of Fourier imaging, the linear relationship between spatial position and the magnetic field would result in a one-to-one mapping between the frequency or phase evolution of a spin's signal and its location, ensuring that a Fourier transform can be performed to resolve the signals into the correct locations. For this mapping to be accurate, the only variations in the magnetic field that the spins in an object should experience would be due to the uniform B_0 main magnetic field and the linear magnetic-field gradients. This uniform background magnetic field is essential for generating an accurate, undistorted image using a Fourier transform reconstruction.

However, several physical effects exist that can disrupt the uniformity of the background magnetic field and, therefore, disrupt the mapping relationship. This can result in misplacement of the signals by the Fourier transform, referred to as image distortion. It can also result in several other artifacts which we will discuss later, including signal loss and deviations in the planned k-space trajectory. Without mitigation and correction, these can result in significant errors in interpretation of an image. Although advances in imaging hardware, sampling strategies, and reconstruction approaches have enabled the reduction of the impact of some of these artifacts, the move to higher magnetic field strengths will increase the importance of considering field inhomogeneity impacts on data acquisitions and image reconstructions. This chapter provides a timely review of the sources and effects of B_0 field inhomogeneity in MRI, as well as existing solutions to address these effects for improved image reconstruction. Although many of the examples presented in this chapter are from brain imaging, the physics models and correction methodology are applicable to other body parts [1–3].

12.2 Disruptions to the homogeneity of the magnetic field

Efforts to create a uniform magnetic field are limited by the magnetic properties of the sample in the magnet. Materials, including tissues in the body, possess a property called magnetic susceptibility, χ . Magnetic susceptibility refers to the magnetizability of a material, and it affects the magnetic field that

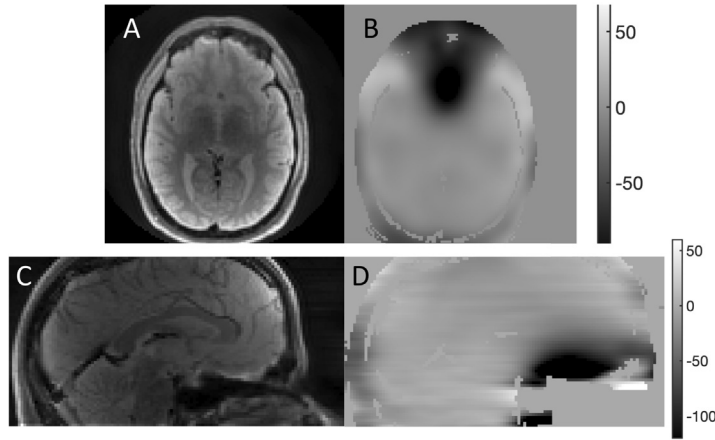
**FIGURE 12.1**

Many areas of functional significance in the brain are near sinuses, resulting in significant field homogeneity disruptions due to large magnetic-susceptibility differences between sinuses and the surrounding brain tissues. Examples of the problematic air–tissue interfaces are shown as a triangle and box at the anterior part of the brain.

a spin experiences in that material, and, consequently, the signals generated. Materials with a positive magnetic susceptibility ($\chi > 0$) are called paramagnetic, and materials with a negative magnetic susceptibility ($\chi < 0$) are called diamagnetic. Magnetic susceptibility shapes the magnetic field around that tissue based on differences between the magnetic susceptibility of surrounding tissues. Of particular importance for brain and body MR imaging is the significant magnetic susceptibility differences between soft tissues ($\chi_{\text{tissue}} = -9 \times 10^{-6}$) and air ($\chi_{\text{air}} = 0.4 \times 10^{-6}$) [4]. Such a difference leads to large field inhomogeneity around the air–tissue interfaces. For example, the sinus spaces around the brain are air-filled and close to interesting regions that are implicated in various pathologies and cognitive functions, as shown in Fig. 12.1. The disruption to the magnetic field’s uniformity can make detection of signals and depiction of accurate functional anatomy in these regions challenging.

12.3 Field inhomogeneity effects on imaging

Given this large magnetic susceptibility difference between air and tissue of almost 10 ppm (parts per million), the magnetic-field uniformity will be significantly disrupted around air–tissue interfaces in the body and the brain. This is shown in Fig. 12.2 for an axial and sagittal image of the brain at a magnetic field strength of 3 T. Although magnetic fields are measured in Tesla, often magnetic field inhomogeneity (or nonuniformity) will be given with units of Hz. Magnetic field and frequency are related through the Larmor equation and the gyromagnetic ratio. If we consider magnetic field

**FIGURE 12.2**

Structural images of the brain and the magnetic field maps corresponding to the same anatomical locations. **A)** An axial brain image showing anatomical structures. **B)** Magnetic field map in Hz measured for the same slice as in A. **C)** A sagittal brain image. **D)** Magnetic field map in Hz at the same slice as in C. Note the large inhomogeneities around the sinuses.

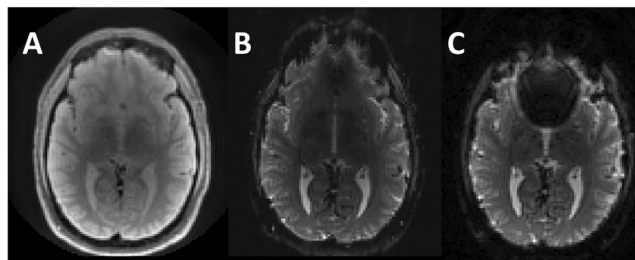
offset (B_{FM} , magnetic field map) to be the deviation from the uniform magnetic field (B_0), such that $B = B_0 + B_{FM}$, then the off-resonance frequency (deviation from the Larmor frequency) can be given as

$$\omega_{FM} = 2\pi f_{FM} = \gamma B_{FM}, \quad (12.1)$$

where ω_{FM} is in rad/sec and f_{FM} is in Hz. When there is a 10-ppm offset in the magnetic field at an air–tissue interface at 3 T (where f_0 is 128 MHz), then we can end up with an off-resonance frequency of 1.28 kHz. For functional MRI (fMRI) applications where the bandwidth per pixel can be on the order of tens of Hz, this can create significant image distortions. An example of the image distortions that can be experienced due to the magnetic field inhomogeneity is shown in Fig. 12.3.

12.3.1 Three types of effects disrupting the image and its information

As already mentioned, there are several impacts of magnetic field inhomogeneity on the imaging process. We will structure our discussion by considering three distinct effects: image distortion, signal loss, and k-space trajectory distortions. Each of these impacts a different part of the image acquisition and reconstruction process and will be discussed separately in this chapter, although, in practice, they are not fully independent. In this section, we will introduce the concept and implications of each of these disruptions. In subsequent sections, we will focus on ways to correct for these artifacts. In addition to these direct effects on imaging and image encoding, which is the focus of this chapter, there are additional impacts of magnetic field inhomogeneity on RF excitation [5,6], spectroscopy [7,8], and other aspects of MRI that will not be discussed (e.g., susceptibility weighted imaging and relaxation effects

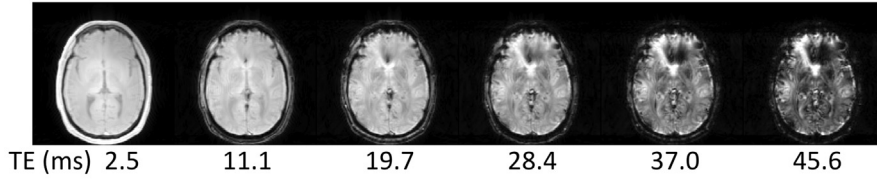
**FIGURE 12.3**

Magnetic field inhomogeneity induced image distortion for the same axial slice in Fig. 12.2. **A**) Structural image showing the true geometry. **B**) EPI image showing the geometric distortion in the phase encode (y-)direction in the region of large field inhomogeneity in the anterior portion of the brain. **C**) Image reconstructed from a spiral acquisition for the same slice showing circular image distortion.

to name a few) [9,10]. In addition, the presence of metal in the body can cause significant and highly localized field inhomogeneity, requiring different approaches [11] that will not be addressed here.

The most commonly encountered and addressed artifact from magnetic field inhomogeneity is image distortion. This artifact results from the disruption to the linear relationship between the magnetic field and spatial coordinates (introduced by linear gradients), resulting in Fourier transform reconstructions to place the signal intensity for a pixel in the wrong place in the reconstructed image. The extent of the image displacement depends on the amount of off-resonance frequency caused by the magnetic-field inhomogeneity, the timing of the acquisition, and the trajectory through k-space [12]. For Cartesian acquisitions, such as echo planar imaging (EPI), the distortion will be primarily in the slow encoding direction, i.e., the phase encode (PE) axis [13]. For non-Cartesian acquisitions, the distortions will be primarily in the direction of the slow traversal through k-space, such as the radial direction for spiral acquisitions [14,15]. It is important to note that this type of distortion exists for all types of imaging acquisitions, including both gradient echo (GRE) and spin echo (SE) acquisitions, and can cause significant errors in subsequent image analysis (e.g., for functional brain activation and network analysis). An example of the image distortion experienced during an EPI acquisition is shown in Fig. 12.3, showing distortion in primarily the PE direction, i.e., the y-direction in the image. The impact and strategies to correct for field inhomogeneity induced image distortions will be addressed in Section 12.4. It is also important to note that chemical shift is another source of off-resonance frequency and will also result in spatial distortions (more discussion can be found in Chapter 14).

A second artifact resulting from field inhomogeneity is intravoxel signal dephasing leading to signal loss since spins within a voxel accumulate phase relative to each other due to different rates of precession from the different magnetic fields experienced. This type of signal loss is prevalent in GRE imaging, where no refocusing pulse is used. The longer the echo time (TE) is used, the more time there is for accumulating phase differences among spins in a voxel and the more signal loss will result, over and above T_2 losses, as shown in Fig. 12.4. In contrast, for SE imaging, the phase that accumulates due to magnetic field inhomogeneity before the refocusing 180° pulse refocuses from the static field inhomogeneity after the refocusing pulse. In the absence of motion of the spins, the field inhomogeneity

**FIGURE 12.4**

Signal dephasing occurs as spins within a voxel are allowed to accumulate different phases, such as when gradient echo images are acquired at longer echo times (TE). Shown is an axial acquisition of a gradient echo image at TE's ranging from 2.5 ms to 45.6 ms with a 4-mm slice thickness at 3 T. Signal loss due to within voxel dephasing can be clearly observed for images at longer TE's.

does not have a large impact. There are a few ways to address the within-voxel signal dephasing effect on the GRE image formation process, and these will be addressed in Section 12.5.

The last artifact due to magnetic field inhomogeneity that will be discussed relates to the magnetic-field gradients that will exist across an object due to the distributions in the susceptibility-induced variations in the background magnetic field. Just as the applied linear magnetic-field gradients are used to traverse k-space in a planned manner for adequate sampling, the background gradients of the magnetic field result in unintended spatial encoding and cause deviations in the planned sampling trajectory, as shown in Fig. 12.5. These deviations are spatially varying and depend on the magnetic field inhomogeneity gradients at each position in the image. These can cause deviations in sample density and field of view that can have significant impacts on the ability to reconstruct an image. K-space trajectory distortions will be discussed further in Section 12.6.

12.3.2 Field inhomogeneity and the signal equation

To rigorously describe the impact of magnetic-field inhomogeneity on the MR imaging process, first we will examine the impact on the signal equation for MR. This formal treatment will provide insights into the various effects, which we will explore in more detail later. The magnetic field inhomogeneity at a voxel will cause spins in that voxel to precess at a different off-set frequency relative to the Larmor frequency defined on the nominal magnetic-field strength. This results in a modification of the signal equation to include this background magnetic-field frequency, $f_{FM}(\mathbf{r})$ in Hz or $\omega_{FM}(\mathbf{r})$ in rad/sec, as

$$s(t_m) = \int_{FOV} x(\mathbf{r}) e^{-i2\pi \mathbf{k}_m \cdot \mathbf{r}} e^{-i2\pi f_{FM}(\mathbf{r}) t_m} d\mathbf{r} \quad (12.2)$$

$$s(t_m) = \int_{FOV} x(\mathbf{r}) e^{-i2\pi \mathbf{k}_m \cdot \mathbf{r}} e^{-i\omega_{FM}(\mathbf{r}) t_m} d\mathbf{r}, \quad (12.3)$$

where $x(\mathbf{r})$ is the continuous transverse magnetization that we are imaging, FOV is the spatial field of view, $\mathbf{r}$ is the spatial variable in 2- or 3-dimensions, $\mathbf{k}$ is the k-space variable in 2- or 3-dimensions, t is the time since the RF pulse for GRE or the center of the refocused echo for SE, m indexes the readout dimension of the data sampling event, and $s(t_m)$ denotes the k-space data acquired at t_m . Note that the time in the readout for a GRE acquisition would be $t_m = TE + m\Delta t$, where TE is the echo time and

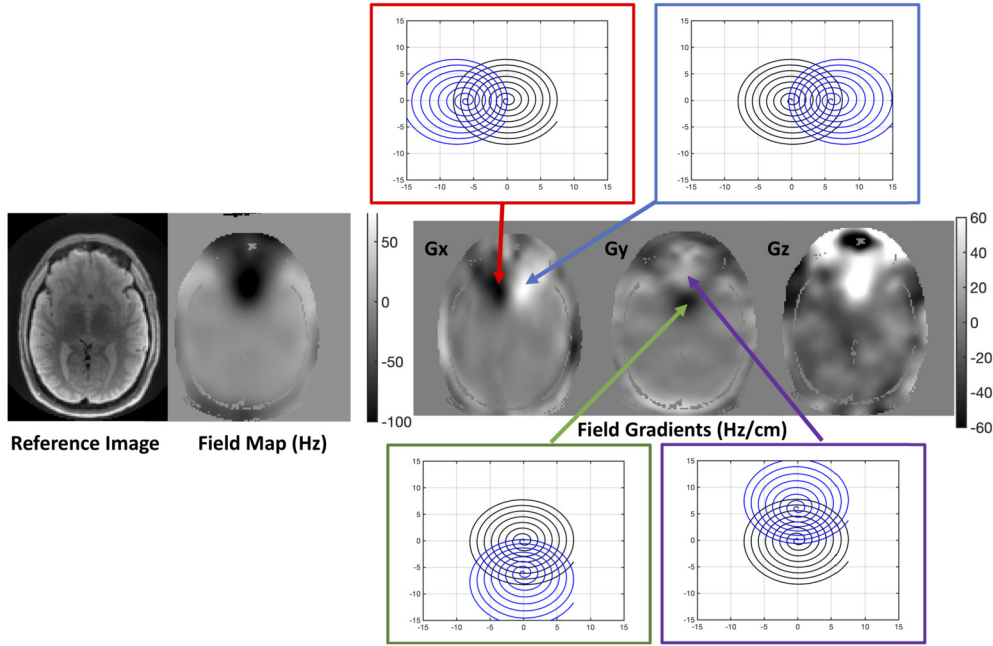


FIGURE 12.5

Anatomical image and magnetic field map in Hz, along with magnetic-field gradients in the x (G_x), y (G_y), and z (G_z) directions. Simulations of spiral trajectories for particular voxels with x- and y-magnetic field gradients are shown in blue (dark gray in print version), compared to the nominal k-space trajectory shown in black. Simulations were performed for ± 30 Hz/cm in x- and y-directions individually.

Δt is the time spacing during the acquisition, and where the bandwidth, BW , of the data acquisition readout is such that $BW = \frac{1}{\Delta t}$.

This equation is formulated as continuous in space, $\mathbf{r}$, but our images will be represented as pixels of constant intensity so we instead approximate our object as a collection of square pixels using the voxel-indicator basis functions, $b_v(\mathbf{r})$, that are centered at each location $\mathbf{r}_n$, for $n = 1, \dots, N$ where N is the number of voxels in the image. The voxel indicator basis functions are small rectangular functions giving a value of 1 within a voxel and 0 elsewhere, as

$$b_v(\mathbf{r} - \mathbf{r}_n) = \begin{cases} 1, & \text{if } \frac{\mathbf{r} - \mathbf{r}_n}{\Delta_r} \leq \frac{1}{2} \\ 0, & \text{else,} \end{cases} \quad (12.4)$$

with Δ_r being the voxel size. Using the voxel indicator (or other suitable basis function), we can express the continuous image as an approximation of coefficients and basis functions, as

$$x(\mathbf{r}) \cong \sum_{n=0}^{N-1} x_n b_v(\mathbf{r} - \mathbf{r}_n). \quad (12.5)$$

In order to get a discrete space model for the signal equation in Eq. (12.3), we must also use a basis expansion of the field inhomogeneity map or field map term. Although representing the image as pixels is straightforward, the expansion basis for the field map will have a significant impact on the signal that is generated by the model. For now, we will expand the field map with the same voxel indicator function as

$$\omega_{FM}(\mathbf{r}) \approx \sum_{n=0}^{N-1} \omega_n b_v(\mathbf{r} - \mathbf{r}_n). \quad (12.6)$$

Using this discrete space model for the spatial basis functions and the field map, we can rewrite Eq. (12.3) as

$$s(t_m) = B_v(\mathbf{k}_m) \sum_{n=0}^{N-1} x_n e^{-i2\pi \mathbf{k}_m \cdot \mathbf{r}_n} e^{-i\omega_n t_m}, \quad (12.7)$$

where $B_v(\mathbf{k})$ is the Fourier transform of the voxel indicator basis function that we used in Eq. (12.4), which is just $B_v(\mathbf{k}) = \text{sinc}(k_x \Delta_x) \text{sinc}(k_y \Delta_y) \text{sinc}(k_z \Delta_z)$ [16].

Now that we have a signal model that incorporates the magnetic-field inhomogeneity or field map, we can evaluate the impact of field inhomogeneity on the image. We can also use our model to enable a variety of reconstruction techniques to estimate images that are free from this distortion. For incorporation of this model into iterative reconstructions, we can embed the field-inhomogeneity terms into the MR encoding operator, $\mathbf{E}$, where the m -th time point and the n -th spatial location impact the encoding operator as follows:

$$\mathbf{E}_{m,n} = B_v(\mathbf{k}_m) e^{-i2\pi \mathbf{k}_m \cdot \mathbf{r}_n} e^{-i\omega_n t_m}, \quad (12.8)$$

such that our signal model becomes

$$\mathbf{s} = \mathbf{E}\mathbf{x} + \boldsymbol{\eta}, \quad (12.9)$$

where $\mathbf{s} = [s(t_1), \dots, s(t_M)]^T$, for M time points in the data acquisition, $\mathbf{x} = [x_1, \dots, x_N]^T$, for N pixels in the image, and $\boldsymbol{\eta} = [\eta_1, \dots, \eta_M]^T$, for the noise at each sampled time point.

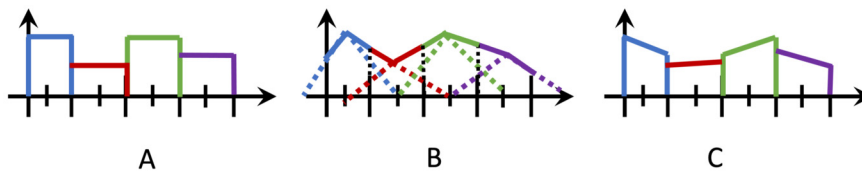
From Eq. (12.7), it is easy to see how image distortion results from unmodeled field inhomogeneity during an acquisition through the accumulation of phase due to field inhomogeneity and the progression of k-space encoding during the acquisition. Consider an EPI trajectory where the k-space trajectory travels consistently through the k_y phase encode axis during the acquisition. In this case, we can find an approximate relationship between k_y and time, such that $t_m \approx ck_m$ in this direction. For a voxel that has a field inhomogeneity value of ω_n , we can look at the encoding kernel which is

$$e^{-i2\pi k_m y_n} e^{-i\omega_n t_m} = e^{-i2\pi k_m y_n} e^{-i\omega_n ck_m} \quad (12.10)$$

$$= e^{-i2\pi k_m y_n} e^{-i2\pi \frac{\omega_n}{2\pi} ck_m} \quad (12.11)$$

$$= e^{-i2\pi k_m (y_n + \frac{c\omega_n}{2\pi})}. \quad (12.12)$$

If a Fourier transform, without field inhomogeneity correction, is used for reconstruction, the spatial representation of the object will be distorted. Image intensity that should have showed up at pixel y_n will

**FIGURE 12.6**

Basis functions that can be used for representing the continuous image and field map for discretization: **A)** voxel indicator basis, **B)** triangle basis functions that create piecewise continuous and piecewise linear representation, **C)** the piecewise linear basis function for which it is computationally convenient within voxel dephasing.

instead appear at a location $y_n + \frac{c\omega_n}{2\pi}$. The magnitude of the spatial misplacement will be $\frac{c\omega_n}{2\pi} = cf_{FM}$. Recall that the constant c is the scaling of a step in sampling time to a step in k-space. Consider the case where c is small, meaning that we are traveling through k-space quickly, the size of the shift in image space will be small. However, for high-resolution EPI acquisitions, c would be larger since it takes more time to sample each line of k-space (one k_y line), and the distortions would also be larger.

The signal model in Eq. (12.7) allows one to achieve effective field inhomogeneity-corrected image reconstruction. But the evaluations of this model is computationally expensive because the Fast Fourier Transform (FFT) can not be directly applied. Methods have been developed to address this computational challenge, and we will discuss some of those in Section 12.4.4.1. These methods use time segmentation or frequency segmentation to break up the computation of the signal equation into several components and add the results back together to approximate the full signal equation. By fixing time or frequency during these sub-evaluations of the signal equation, they can leverage the FFT to achieve faster computations.

12.3.2.1 Other basis expansions enable the modeling of additional artifacts

In Eq. (12.4), we used a voxel indicator's basis function to transform our continuous space signal equation into a discrete space signal equation focused on reconstructing an image of voxels with one coefficient per uniform voxel. Although this makes sense for a representation of the image, magnetic-field inhomogeneity can cause signal dephasing due to a distribution of the magnetic field within a voxel. A voxel indicator basis function does not enable us to model variations in the magnetic field within a voxel. To accommodate this within-voxel dephasing, we need to represent the magnetic-field inhomogeneity with a basis function that varies within a voxel.

Ideal expansions would result in continuous functions of the magnetic-field distribution to have a realistic approximation. However, the overlapping basis functions, such as triangle basis functions, can cause mathematical difficulties in trying to incorporate such models into efficient image-reconstruction routines [16,17]. So, to maintain convenience of computation, we can choose piece-wise linear basis functions that have a mean value and linear slope for each voxel. This results in piecewise linear approximations of the field distribution, but not a continuous function. The various bases discussed are shown in Fig. 12.6, including the voxel indicator function, triangle basis functions, and piecewise linear basis functions.

The piecewise linear basis expansion for the field map requires both a mean field-inhomogeneity value, ω_n for each of the N pixels, but it also requires a slope of the field map in each direction, $\mathbf{G}_n = [X_n, Y_n, Z_n]^T$ which are measured in the x -, y -, and z -direction in rad/s/pixel [17,18]. This gives the following approximations for the image and the field map, where $b_v(\mathbf{r})$ is the same voxel indicator function we used before,

$$x(\mathbf{r}) \approx \sum_{n=0}^{N-1} x_n b_v(\mathbf{r} - \mathbf{r}_n), \quad (12.13)$$

$$\omega_{FM}(\mathbf{r}) \approx \sum_{n=0}^{N-1} \left(\omega_n + \mathbf{G}_n^T \cdot \frac{\mathbf{r} - \mathbf{r}_n}{\Delta_r} \right) b_v(\mathbf{r} - \mathbf{r}_n). \quad (12.14)$$

If we integrate this approximation into the continuous signal equation in Eq. (12.3), we end up with the following signal model:

$$s(t_m) = \sum_{n=0}^{N-1} B_v(\mathbf{k}_m, \mathbf{G}_n, t_m) x_n e^{-i2\pi \mathbf{k}_m \cdot \mathbf{r}_n} e^{-i\omega_n t_m}, \quad (12.15)$$

where the function B_v is now a more complicated Fourier transform of the voxel indicator function that includes the field inhomogeneity gradient terms. This is given as

$$B_v(\mathbf{k}_m, \mathbf{G}_n, t_m) = \text{sinc} \left(k_{x,m} \Delta_x + \frac{X_n t_m}{2\pi} \right) \text{sinc} \left(k_{y,m} \Delta_y + \frac{Y_n t_m}{2\pi} \right) \text{sinc} \left(k_{z,m} \Delta_z + \frac{Z_n t_m}{2\pi} \right). \quad (12.16)$$

It is important to consider the units of all the terms in this signal equation model when implementing it, including the units of k -space, the field map, the field map gradients, and the voxel size.

We can update our encoding operator, $\mathbf{E}$, to include this advanced signal model. Entries of $\mathbf{E}$ now become

$$\mathbf{E}_{m,n} = B_v(\mathbf{k}_m, \mathbf{G}_n, t_m) e^{-i2\pi \mathbf{k}_m \cdot \mathbf{r}_n} e^{-i\omega_n t_m}. \quad (12.17)$$

Note that the $B_v(\mathbf{k}_m, \mathbf{G}_n, t_m)$ term is computationally expensive to compute as it now depends on both spatial position and time (k -space), hence, we may ask: Why is it still inside the summation in Eq. (12.15) compared to Eq. (12.7). This formulation again precludes the use of FFT for evaluation and requires the use of explicit calculation of the signal equation. Recently, methods have been developed to approximate the full encoding matrix with a low-rank approximation [19,20]. These approximations enable speedups in computation of several orders of magnitude, making this more complete signal model feasible to use in iterative reconstruction schemes. We will discuss the image reconstruction approach that incorporates this model further in Section 12.5.1.

By modeling the linear components of the variations in magnetic field within a voxel, this signal model enables us to address some of the signal dephasing that occurs during imaging. As an example of that, consider 2D gradient echo imaging. For this case, there is no k_z encoding and the $B_v(\mathbf{k}_m, \mathbf{G}_n, t_m)$

term reduces to

$$B_v(\mathbf{k}_m, \mathbf{G}_n, t_m) = \text{sinc}\left(k_{x,m}\Delta_x + \frac{X_n t_m}{2\pi}\right) \text{sinc}\left(k_{y,m}\Delta_y + \frac{Y_n t_m}{2\pi}\right) \text{sinc}\left(\frac{Z_n t_m}{2\pi}\right). \quad (12.18)$$

Focusing only on the z-direction, i.e., the direction through the imaging slice and the through-plane direction, we can see that the signal will decay as a sinc() function with increasing decay as the through-plane gradient and time increase. If there were no in-plane components (x - and y -directed gradients in the magnetic field map), then the signal would exhibit the through-plane signal dephasing causing signal loss that can not be recovered with a Fourier reconstruction, especially for gradient echo acquisitions with long echo times.

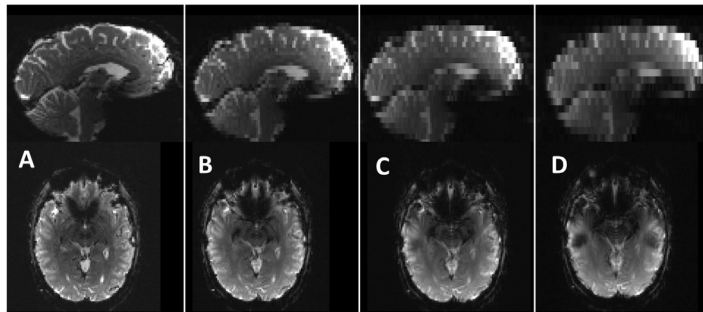
12.3.3 Field inhomogeneity mitigation methods

Before leaving the description of the basic physics of magnetic-field inhomogeneity, we should note that there are several ways to decrease the impact of field inhomogeneity on the image. These can be through hardware-based approaches that minimize the magnitude of the field inhomogeneity distribution over the sample to be imaged. Or they can recognize the dependencies in the signal equation and choose imaging parameters that result in smaller artifacts.

From the early days of functional MRI of the brain, MR engineers realized that the best way to minimize the impact of field inhomogeneity was to minimize the amount of field inhomogeneity in the first place. This can be helped by careful shimming procedures and the use of higher-order shims that are common in modern systems [21–23]. Additionally, the magnetic field distribution can be impacted by placing items in and around the head to reduce the field variations, especially near the orbitofrontal region of the brain. These items included the development of a dielectric mouth insert [24–26], a shim coil that was placed in the mouth [27], or dielectric foam [28]. This has progressed more recently to the use of multichannel receiver coils that put DC currents through the localized loops to create magnetic fields that counteract the local magnetic-field inhomogeneities induced in the sample due to magnetic susceptibility [29]. As receiver coils with higher channels are built, these enable a customized shimming for every sample in the scanner. A review of shimming methods, targeting 7 T MRI systems and the brain, describes several other approaches [30].

Once the magnetic field has been made as uniform as possible, there are several imaging parameters that can have an impact on the amount of distortion or artifacts that will be experienced. As demonstrated in Section 12.3.2, the timing of the acquisition and its relationship to the k -space trajectory can be an important determinant of the size of the artifact. Faster imaging due to higher-bandwidth acquisitions, larger gradient amplitudes, and parallel imaging can reduce the size of the image distortion experienced. These improvements in image fidelity often come with a cost of signal-to-noise ratio, resulting in a tradeoff. For through-plane dephasing, it is the distribution of magnetic fields across the voxel that result in signal cancellation. Thus, thinner imaging slices (smaller voxels) will directly reduce this range of frequencies, and hence, the dephasing in the slice [31–33]. Fig. 12.7 shows a 2D EPI functional MRI acquisition performed at 3 T with various slice thicknesses to illustrate this effect.

In addition to optimizing the parameters, some simple modifications to a pulse sequence can also result in reducing the impact of susceptibility-induced artifacts. One of these is to dynamically update the shim gradients and receive frequency on a slice-by-slice basis to counteract linear components of the field inhomogeneity distribution [34]. Similarly, Z-shimming is the use of an extra gradient

**FIGURE 12.7**

Decreasing the slice thickness reduces the within voxel dephasing effect. EPI acquisitions from an fMRI scan with slice thickness of **A)** 2 mm, **B)** 4 mm, **C)** 6 mm and **D)** 8 mm.

across a slice (Z-direction) to counteract the dephasing that occurs due to magnetic-field inhomogeneity gradients through the slice direction (which is usually the direction with stronger field gradients or larger dimension) [35–38]. Early methods were limited to only one z-shimming per slice, although several single-shot methods were developed to perform a few z-shimmings in a single-shot acquisition [39–42]. Modern sequences overcome these limitations by designing fully shaped RF pulses that apply a spatial map of through-plane phase, providing compensatory phase to every region of the brain as needed [43,5,44,45]. These RF pulses may become quite long due to the need to provide high-resolution spatial phase to a slice, but the development of multiple channel transmit systems enables time-efficient RF pulses to be used.

Despite these mitigation approaches, field-inhomogeneity effects persist in the image acquisition. Left uncorrected, these will lead to imaging artifacts that can misplace signals, to improper contrast, or to inadequate k-space sampling. In the following sections, we will focus on modeling and correcting these field-inhomogeneity effects.

12.4 Image distortions and correction approaches

As previously noted, image distortion affects nearly all types of image acquisition approaches including both spin echo and gradient echo. Therefore, it is important to understand the impact, how to minimize the distortion, and how to correct for it during or after image reconstruction.

12.4.1 Distortions depend on trajectory and sample timing

As we saw previously, the amount of distortion depends on the magnitude of the magnetic-field offset and the timing of the acquisition trajectory. A useful concept to enable easy interpretation of the impact of field inhomogeneity on an acquisition is the concept of bandwidth per pixel (BWPP). When imaging, if a gradient is applied across a certain direction, for example the x -direction, then we get a distribution of frequencies of spins across this direction. When sampling with the Nyquist criteria, the individual

pixels are represented as bins of frequencies within a certain range. Assuming that the correct sampling rate and the correct gradient amplitude have been chosen, the total bandwidth of the acquisition is determined by the inverse of the sample time spacing, $BW = \frac{1}{\Delta t}$. If we consider that the range of frequencies will be spaced across the N pixels of the field of view, then we can calculate the BWPP as

$$BWPP = \frac{BW}{N} = \frac{1}{N \Delta t} \quad (12.19)$$

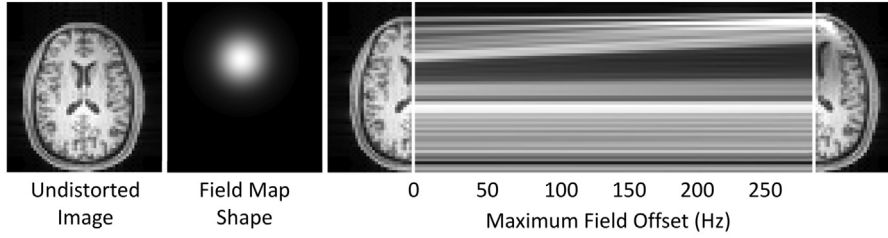
in Hz per pixel. This interpretation makes it easy to figure out how many pixels of distortion that a particular field inhomogeneity, measured in Hz, will create with a Fourier transform reconstruction. It also makes it easy to determine how choices in the acquisition will impact this level of distortion. A faster time sampling will result in a larger BWPP and lower image distortions (i.e., a smaller c in Eq. (12.12)).

The previous example was for a simple gradient readout event across a single axis, the readout axis, which would be the situation for a Cartesian acquisition with a single readout per TR. In this scenario, typically experienced magnetic field-inhomogeneity levels would not result in measurable image distortion due to the high BW in the data acquisition. If the data sampling of an MRI scanner is 200 KHz, i.e., a time spacing of 5 μ s, then, for an image matrix size of $N = 200$, the BWPP would be 1 KHz. Given that, at 3 T, the maximum magnetic susceptibility difference between air and tissue is about 1 KHz, we would expect to get a distortion of, at most, 1 pixel across the 200 pixel field of view. This would have a minor impact on the spatial integrity of the image. However, this concept of BWPP works in the other encoding directions, too. For an EPI acquisition, the data sampling along the phase encode direction is not continuous, but instead has a sampling that is spaced by a larger amount of time, called the echospacing (T_{Echosp}). The echospacing is slightly larger than the time required to measure a whole line in the readout direction. We can think of the sampling bandwidth in this PE direction as $BW_{PE} = 1/T_{Echosp}$, and get the BWPP in the PE direction as

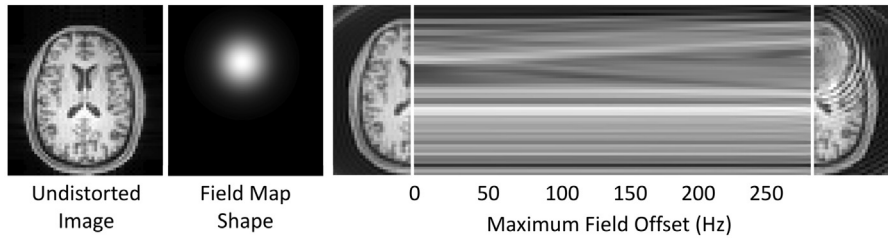
$$BWPP_{PE} = \frac{BW_{PE}}{N_{PE}} = \frac{1}{T_{Echosp} N_{PE}}, \quad (12.20)$$

where N_{PE} is the number of pixels in the phase encode direction. For an EPI acquisition with an echospacing of ≈ 0.4 ms and 64 pixels, the $BWPP_{PE}$ would be 39 Hz/pixel. In this case, the fat–water frequency difference at 3 T (≈ 440 Hz) would result in more than 11 pixels displacement in a 64-pixel image that would be a very significant image distortion.

To demonstrate the dependence of the spatial shift in EPI on the magnetic-field inhomogeneity, in Fig. 12.8 a single vertical line of the reconstruction of a brain image is shown with increasing magnetic-field inhomogeneity. This more general formulation of BWPP can be used to approximate the size of the spatial distortion with other k-space trajectories. However, the distortions from magnetic-field inhomogeneity may not be simple geometric shifts when trajectories are used in which the k-space and the time axis are not just scaled versions of each other. For example, in spiral trajectories, such as the spiral-out trajectory, the k-space trajectory starts at the center of k-space and increases its radius as time progresses in the sampling. This means that the azimuthal sampling is very fast and would have a high bandwidth, but the radial sampling is slower, requiring the sampling trajectory to make a full turn through k-space to get to the next sample in the radial direction. The spiral trajectory and its distortion are demonstrated in Fig. 12.9. The radial spacing may not be exactly evenly spaced in time as the spiral

**FIGURE 12.8**

Distortion of EPI acquisition as a function of size of the magnetic-field inhomogeneity for an acquisition with 64 pixels and an echospacing of 0.4 ms. Undistorted image and the magnetic-field map shape along with an image showing the distortion of the images since the maximum magnetic field map value goes from 0 Hz to 280 Hz. The x-axis shows increasing field inhomogeneity in Hz. The y-axis shows one line along the y-axis (PE axis) in the image, depicting the distortion.

**FIGURE 12.9**

Distortion of spiral out acquisition with increasing magnetic-field inhomogeneity for a single-shot spiral acquisition with a 20-ms readout for a 64 matrix size acquisition. Similar to Fig. 12.8, the right-most image shows the distortion for the midline of the image as the maximum-field inhomogeneity goes from 0 to 280 Hz.

progresses from smaller inner rungs to outer rungs, so the approximation involved in a $BWPP_{radial}$ is only approximately true. If we assume an $N = 64$ matrix size spiral takes $T_{dur} = 20$ ms to acquire and includes 32 ($N/2$) rungs, then we can approximate the radial bandwidth as, $BW_{radial} = 1/T_{rung}$, where $T_{rung} = T_{dur}/(N/2) = 2T_{dur}/N$. So, $BWPP_{radial}$ is given by

$$BWPP_{radial} = \frac{BW_{radial}}{N} = \frac{1}{T_{rung}N} = \frac{N}{2T_{dur}N} = \frac{1}{2T_{dur}}. \quad (12.21)$$

So, the $BWPP_{radial}$ is approximately the inverse of twice the spiral readout duration. In this case, that would be 25 Hz/pixel. So, the fat–water frequency shift at 3 T would result in a 17.6 pixel radial blur. This radial blur would extend in all radial directions from the original location of the signal experiencing this field inhomogeneity, resulting in an overall spread of the point spread function of a signal to 35 pixels out of 64. A signal distortion this large would create significant artifacts in images without fat saturation pulses. Partial mitigation of this effect is a necessity and can be attempted through fat saturation pulses, water–fat separation techniques, multishot acquisitions (reducing spiral readout duration

per shot through the use of multiple shots) and parallel imaging (which also decreases spiral readout duration). Fig. 12.9 shows the image distortion for a pixel placed in a magnetic-field inhomogeneity of varying intensity. The magnitude of field inhomogeneity is shown on the x-axis, while a single line through the image is shown along the y-axis. Note that the impact of distortions would be similar along both the x- and y-axis of the image for a spiral-out acquisition.

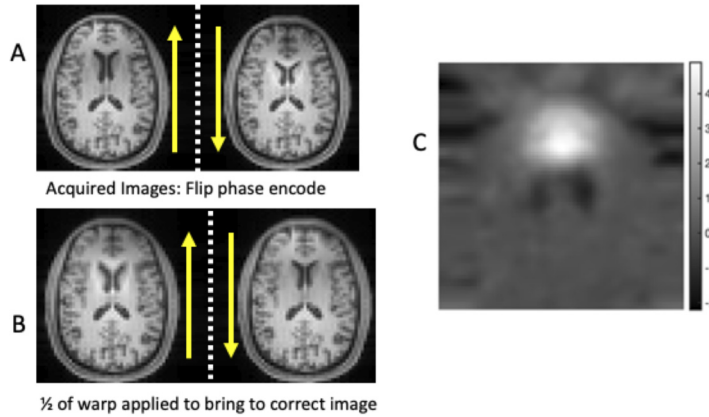
12.4.2 Image correction: image warping approaches

Given that even moderate magnetic-field inhomogeneity can result in substantial spatial distortion in the image, there have been many methods developed to address this distortion during image reconstruction. We start with a simple approach that is commonly used for EPI, where the distortion is primarily in one direction and represented as a spatial shift. We saw previously that field inhomogeneity results in spatial shifts in the encoding axis that is slow in time (hence, low in BW), i.e., the phase-encode direction. Correction for the geometric shift induced by this shift can be corrected through image registration-based approaches that acquire two acquisitions with the phase-encoding direction (and, hence, the axis of the spatial shift) flipped [46,47]. With one acquisition acquired with the PE direction anterior–posterior and another acquired with the PE direction posterior–anterior, the spatial shifts between the two images will be in opposite directions. Nonlinear spatial warping can be done to align one of the images to the other. By applying half the spatial warp to each image, the resulting image will be undistorted. The spatial warp field can be converted to a traditional magnetic field-inhomogeneity map if desired by using the sequence timing to convert the warp field for each voxel back to magnetic-field offset. The disadvantage of this approach would be the need to acquire two images to estimate one undistorted image. However, both fMRI and DTI acquire many EPI images that are subject to the relatively constant magnetic-field inhomogeneity. So, typically, an extra acquisition is acquired before the fMRI or DTI sequence (thus, with minimal time penalty), usually with a spin echo EPI to minimize through-plane signal loss, with the flipped phase-encode images. The distortion map or field map estimated from this flipped PE axis pair is used to correct all subsequent images. Fig. 12.10 shows an example of flipping the phase-encode axis to estimate the magnetic field map and undistorted images. This is the approach that was taken in the Human Connectome Project to provide time-efficient field-inhomogeneity correction to the EPI-based scans (fMRI and DTI) [48].

This method works only for acquisitions, such as EPI where the image distortion manifests as a pixel shift. There are many tools that enable this to be accomplished in a straightforward manner, such as *topup* in FSL [49,50], with just a few parameters that are needed from the acquisition of the flipped PE axis scans and the fMRI or DTI run that is to be corrected. Further advances to this method incorporate spatial atlasing or other anatomically constrained distortion estimation based on other images acquired during a full imaging study and not necessarily a differentially distorted (flipped phase-encode direction) EPI image [51,52].

12.4.3 Image correction: conjugate phase

The image warping approach just explained is limited to trajectories, such as EPI, where there is a linear relationship between the slow k-space axis and time evolution during the acquisition. This results in distortions that are simple geometric shifts. For k-space trajectories that do not fit this acquisition framework, a more direct incorporation of the magnetic field-inhomogeneity map is required in the

**FIGURE 12.10**

Flipping the phase encode axis to find the field inhomogeneity-induced distortion. **A)** Images acquired with EPI with opposite PE directions (y-axis), showing image distortions in opposite directions. **B)** After estimating the warping field, half of the warp is applied to each image to bring them back together within an undistorted image. **C)** The warping field is displayed as a map of pixel shifts in the PE direction (simulated).

image reconstruction step. This approach is required for spiral, radial, and rosette trajectories, but also for EPI that do not have the simple linear relationship, such as multishot and centric-reordered EPI.

When incorporating the magnetic field-inhomogeneity map into the image reconstruction, you must first have a measurement of the magnetic-field map, i.e., the spatial distribution of the magnetic field inside the sample that you are imaging. Methods for measuring the field map will be discussed in Section 12.7. For now, we assume that an accurate field map, $\omega_{FM}(\mathbf{r})$, is available.

Once we have an estimate of the magnetic field-inhomogeneity map, we can proceed to invert (or approximately invert) Eq. (12.3) to create a field-corrected image. Since the signal in a standard MRI acquisition is just the Fourier transform of the image, at a base level, the image is just the inverse Fourier transform of the data. However, we must refocus the phase effects of the magnetic-field inhomogeneity in order to bring the net encoding (k-space and field inhomogeneity) of the signal from a pixel back into focus. This approach is called the conjugate-phase method since we apply the conjugate of the phase accumulation due to the magnetic-field inhomogeneity back to the acquired data as we reconstruct a pixel [53–55]. The continuous formulation of this is given as

$$\hat{x}(\mathbf{r}) = \int_{Wk} s(\mathbf{k}(t)) e^{i2\pi\mathbf{k}\cdot\mathbf{r}} e^{i\omega_{FM}(\mathbf{r})t} d\mathbf{k}, \quad (12.22)$$

where Wk indicates that the integral is performed over all of continuous k-space. Converting this equation into our case of sampled k-space data requires several considerations, including the basis function expansion of the image discussed in Section 12.3.2 and consideration of the sample density-compensation function, $H(\mathbf{k})$, that takes into account the nonuniformly sampled k-space during an acquisition such as a spiral trajectory. If we assume a voxel-indicator basis function, as was used in Eq. (12.4), and we lump the k-space scaling from the basis function into the sample density com-

pensation function, we get the following image reconstruction equation that corresponds to the signal equation in Eq. (12.7), as

$$\hat{x}(\mathbf{r}_n) = \sum_{m=0}^{M-1} H_m s(t_m) e^{i2\pi \mathbf{k}_m \cdot \mathbf{r}_n} e^{i\omega_n t_m}, \quad (12.23)$$

where H_m includes sample density compensation and division by $B_v(\mathbf{k})$. If we let $\mathbf{H} = [H_0 \dots H_{M-1}]^T$, then we can reformulate the conjugate-phase reconstruction in a similar way to the encoding operator in Eq. (12.9) as

$$\hat{x} = \mathbf{E}^H (\mathbf{H} \circ \mathbf{s}), \quad (12.24)$$

where $\circ$ is the Hadamard or element-wise product of the data and weighting vectors and $[]^H$ is the Hermitian transpose, which takes the complex conjugate transpose of the encoding matrix, including the k-space encoding and the field-inhomogeneity term.

Calculation of Eq. (12.23) can be computationally demanding since the kernel that needs to be computed depends on both spatial position, $\mathbf{r}_n$, and time, t_m . This cannot be computed by a FFT directly, and, instead, direct calculation of this reconstruction would require full (and slow) computation of the matrix–vector product of $\mathbf{E}^H$ and $\mathbf{s}$. It should be noted that explicit storing of the large matrix $\mathbf{E}$ must also be avoided since it can be too large to store in computer memory for even moderately sized problems. Instead of direct calculation of the matrix–vector product, we can leverage several FFT’s by decomposing the problem into several components by performing either time segmentation [54] or frequency segmentation [55]. With time segmentation and the use of a small time interval, we can drop the time dependence out of the kernel term: $e^{i\omega_n t_m}$ and incorporate field inhomogeneity-induced phase only in the end points of the time interval. If we break up the readout into L time segments, each τ seconds long, we get the following for the reconstruction using an interpolating window function $\alpha(t)$ such as a Hanning window [54]:

$$\hat{x}_n = \sum_{l=0}^L e^{i\omega_n l\tau} \sum_{m=0}^{M-1} \alpha(t_m - l\tau) H_m s(t_m) e^{i2\pi \mathbf{k}_m \cdot \mathbf{r}_n}. \quad (12.25)$$

The second summation is just a weighted reconstruction of the k-space data and can be accomplished quickly with gridding for non-Cartesian k-space trajectories [56].

12.4.4 Image correction: inverse problem approach

The conjugate-phase approach provides a one-step, direct reconstruction of magnetic field inhomogeneity-corrected imaging data from arbitrary k-space trajectory acquisitions. This approach works well for fully sampled k-space conditions with a smooth magnetic field-inhomogeneity map. We will discuss comparisons and limitations later in Section 12.4.5. However, in many imaging situations, the data are undersampled and in need of parallel imaging, and additional prior information is available and/or compressed sensing reconstructions are needed in order to produce a high-quality image. For these cases, it is necessary to formulate the MRI image-reconstruction problem in an inverse problem approach. In

addition to these cases, the inverse problem approach is not limited by some of the approximations required for a one-step direct reconstruction like the conjugate-phase reconstruction. Conjugate-phase approximations include the need for accurate sample density compensation, $H(\mathbf{k})$, which we will show is potentially complicated by magnetic-field inhomogeneity in Section 12.6. Inverse problem approaches, by simulating the acquired signal, also provide energy-preserving reconstructions in the presence of regions with a high slope in the magnetic field-inhomogeneity map, which will be discussed further in Section 12.4.5.

For the inverse problem approach, additional basis functions can be easily incorporated into the signal model, as was discussed in Section 12.3.2.1. We formulate the image reconstruction using the signal equation model that was given in Eq. (12.9). Then the image reconstruction progresses by finding the image $\mathbf{x}$ that results in a signal that best matches the acquired data, $\mathbf{s}$, as discussed previously in Chapter 2. This optimization problem can be formulated in its simplest form as a least squares problem as

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{s} - \mathbf{E}\mathbf{x}\|^2, \quad (12.26)$$

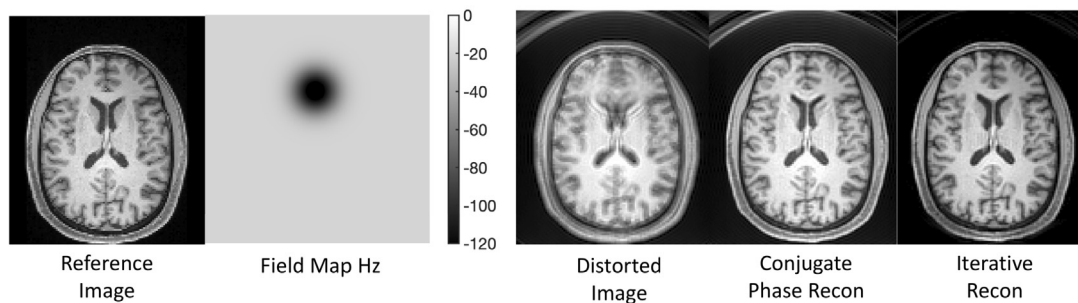
where we have only included a data fit term. Or, it could incorporate additional prior constraints on the image $\mathbf{x}$, along with coil sensitivity information for a SENSE-type parallel imaging [57] (see also Chapter 6). Several open-source implementations of this type of image reconstruction are readily available.¹ This model can be easily incorporated into algorithms covered in other chapters of this book, including constrained, low-rank, and dictionary-based approaches.

12.4.4.1 Computational considerations

The inverse problem approach results in an iterative image-reconstruction algorithm that can take significant time to compute due to the term $e^{-i\omega_{ntm}}$ that depends on both spatial position and time. Significant computational speedups can be achieved using similar approaches to the time segmentation that was mentioned for conjugate phase. However, efficient computation is even more critical in iterative reconstruction since the field inhomogeneity-map term will be calculated many times. Additional optimization has been performed to reduce the number of time segments required to get an accurate approximation using a min-max optimal interpolator [16].

Iterative reconstructions must simulate the complicated signal models of $\mathbf{E}$ and $\mathbf{E}^H$ multiple times during image reconstruction. For field inhomogeneity-corrected image reconstructions, due to time segmentation, this usually results in an order-of-magnitude increase in computation time. In order to make field inhomogeneity corrected reconstruction feasible, several other approaches to speedup reconstruction have been developed, including those leveraging graphics processing units (GPUs) and other high-performance computing approaches [58–61]. Through these speedups, work is being done to try to make the field inhomogeneity-corrected reconstructions work in clinically feasible times, which would enable advanced non-Cartesian trajectories with time-efficient, long data readouts usable for clinical imaging purposes.

¹ Please see [16] and associated code included in the Michigan Image Reconstruction Toolbox (MIRT, <https://web.eecs.umich.edu/~fessler/code/>) or the Reconstruction code from Illinois (RecoIL, <https://github.com/mrfil/RecoIL>) as examples.

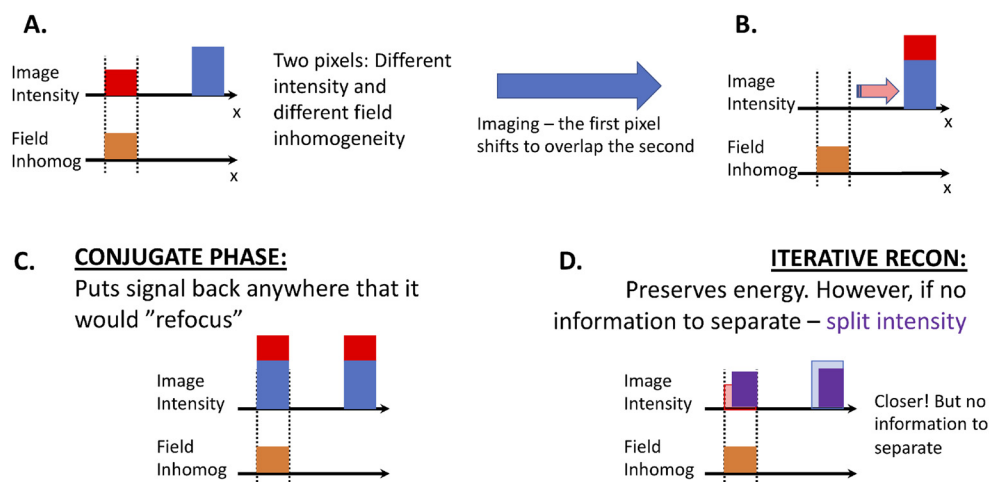
**FIGURE 12.11**

An axial slice of the brain (reference image), a corresponding field map (in Hz), and reconstructions obtained from a simulated single-shot spiral acquisition with a matrix size of 128 acquired with 2-shot acquisition (equivalent to a single-shot parallel imaging factor of 2 in terms of distortion). The distorted image from a Fourier transform reconstruction shows blurring, but the images reconstructed with both time-segmented conjugate phase and time-segmented NUFFT-based iterative reconstruction approach (10 iterations) recover a sharp image.

12.4.5 Comparing performance of image correction approaches

An example image simulated with magnetic field inhomogeneity and the two reconstruction methods of conjugate-phase and iterative reconstruction are shown in Fig. 12.11. As demonstrated in this example, magnetic field correction methods drastically improve image quality. However, the iterative reconstruction method provides a more complete correction for the field-inhomogeneity effects.

In principle, the conjugate-phase approach refocuses signal intensity to the region where it originated, as indicated by the encoding of k -space and the magnetic-field inhomogeneity it experiences. However, there are many imaging situations where there may be ambiguity about the origin of signal. Such a situation would arise with an EPI-like trajectory, causing a geometric shift of a pixel with high-field inhomogeneity (pixel 1) onto a different location with very low-field inhomogeneity (pixel 2). This situation is demonstrated schematically in Fig. 12.12 for a theoretical 1D imaging experiment. If an uncorrected image is formed, all signal intensity shows up in the location of pixel 2, i.e., it appears that all the signal is coming from pixel 2. The signal encoding (k -space/gradients) combined with the field-inhomogeneity information still does not provide sufficient information to distinguish whether the signal intensity originated from pixel 1 or pixel 2. Conjugate-phase and iterative reconstructions handle this situation differently, as demonstrated in the figure. For conjugate-phase reconstruction, the signal will refocus at both positions, and the conjugate phase reconstruction will put the full amount of signal (sum of pixel 1 and pixel 2) into both locations. The conjugate-phase reconstruction is not energy preserving since it will refocus signal in every location from which it could have originated. In contrast, the iterative reconstruction simulates the signal and tries to match the simulated signal to the received signal. This must be energy preserving in the image and signal domain. So, in the absence of other prior information, it will split the signal between both locations. Although this is closer to the correct image in terms of error, it does not contain any additional information about the actual signal at pixel 1 and pixel 2. However, if prior information is available from surrounding pixels and a regularization

**FIGURE 12.12**

A simplified 1D example of the different behavior of the conjugate phase and iterative reconstruction methods with respect to placing signal intensity with ambiguity due to magnetic-field inhomogeneity. **A).** The setup of the problem where two pixels (red and blue (mid gray and dark gray in print version)) have different intensity, but the red pixel also suffers from field inhomogeneity shifting it, in the distorted image (panel **B**), onto the same location as the blue pixel. **C).** Since the signal intensity refocuses to both locations, conjugate phase will place the sum of both pixel intensities at both locations. **D).** The iterative reconstruction is energy preserving and will split the signal between the two locations if no other prior information is available to help determine how much of the signal came from each location.

approach is used in the iterative reconstruction method, then a more accurate image can be formed by finding the best fit image that follows the prior information.

12.5 Phase and signal dephasing correction approaches

In addition to causing geometric distortion, magnetic-field inhomogeneity can lead to phase accumulation due to off-resonant spins. This occurs mainly in gradient echo imaging where the phase is allowed to accumulate over time before or during the acquisition. This phase accumulation can lead to errors in extracting information from images that are dependent on the phase, such as in magnetic resonance elastography [62,63]. Additionally, it can lead to signal loss within a voxel if there are variations in the magnetic field across that voxel such that spins experience a range of fields within that voxel. These spins within a voxel have various frequencies and, over time, they will start to cancel one another, leading to signal loss. This signal dephasing can result in images with large regions of holes or signal voids, especially in functional MRI near the orbitofrontal cortex or around the ear canals. Although other regions may not show complete signal voids, the loss of signal due to magnetic field inhomogeneity-induced dephasing can still cause problems, especially in quantitative imaging methods.

There are several ways to mitigate this effect prior to the image-reconstruction step. This includes the z-shimming and shaped RF pulses that were mentioned in Section 12.3.3. In addition, these gradients through a voxel can be interpreted as additional linear gradient-encoding functions resulting in a unique k-space trajectory for each voxel. This will be discussed in Section 12.6 and can lead to additional mitigation strategies, such as using full 3D k-space sampling or sampling additional locations in k-space to ensure adequate coverage.

12.5.1 Image reconstruction based approaches for within voxel dephasing

The signal loss caused by within voxel dephasing can produce significant artifacts in the image. This is particularly important for gradient echo imaging, multi-echo imaging, spectroscopic imaging, and functional MRI if the voxel size is not small enough to ignore intravoxel B_0 variations. In order to address these effects during the image reconstruction step, we must use a signal model that includes the field gradients. As mentioned in Section 12.3.2.1, we can impose a piecewise linear model of these within-voxel gradients as in Eq. (12.14), leading to a signal model as in Eq. (12.15). This model has a term, $B_v(\mathbf{k}_m, \mathbf{G}_n, t_m)$, that depends on both spatial position and k-space sampling locations that causes significant computational and memory challenges for incorporating this correction into reconstruction algorithms. To this end, post-processing methods have been described to correct the effects of intravoxel B_0 [64–67, 9, 1, 68]. However, these methods require explicit knowledge of the point spread functions (PSFs) and can become intractable if advanced reconstruction methods are used.

Fortunately, recent methods have been developed to make computation related to $B_v(\cdot)$ more efficient through low-rank approximations [19, 20] (see Chapter 9 for more discussions on low-rank modeling). Specifically, $B_v(\mathbf{k}_m, \mathbf{G}_n, t_m)$ can yield an accurate partially separable functions-based approximation, i.e.,

$$B_v(\mathbf{k}_m, \mathbf{G}_n, t_m) \approx \sum_{l=1}^L \mathbf{u}_l(\mathbf{k}_m) \mathbf{v}_l^T(\mathbf{G}_n), \quad L \ll M, N. \quad (12.27)$$

If we express the entire operator using a matrix form $\mathbf{\Omega}_v \in \mathbb{R}^{M \times N}$, Eq. (12.27) implies that $\mathbf{\Omega}_v$ has a low-rank representation $\mathbf{\Omega}_v = \mathbf{U}\mathbf{V}$, with $\mathbf{U} \in \mathbb{R}^{M \times L}$ and $\mathbf{V} \in \mathbb{R}^{L \times N}$ being two rank- L matrices. Importantly, $\mathbf{U}$ and $\mathbf{V}$ can be estimated without explicitly forming $\mathbf{\Omega}_v$ in the process. Specifically, a subset of the columns in $\mathbf{\Omega}_v$ can be formed by evaluating Eq. (12.16) at the sampled k -space coordinates and a random subset of voxel field gradients calculated from a pre-estimated B_0 map. SVD can then be applied to this column subset to estimate $\mathbf{U}$. With $\mathbf{U}$ estimated, $\mathbf{V}$ can be efficiently estimated by a least-squares fitting using a row subset of $\mathbf{\Omega}_v$ formed similarly as previously described, but with a subset of k -space coordinates instead. This allows us to decompose the complicated encoding operator $\mathbf{E}$ into L components as

$$\mathbf{E}\mathbf{x} \approx \mathbf{B} \sum_{l=1}^L \mathbf{u}_l \circ (\mathbf{F}\mathbf{v}_l \circ \mathbf{x}), \quad (12.28)$$

each of which contains a voxel-wise multiplication between the image of interest and $\mathbf{v}_l$, a Fourier transform operator $\mathbf{F}$ that can be easily implemented using FFT's, a k -space point-wise multiplication

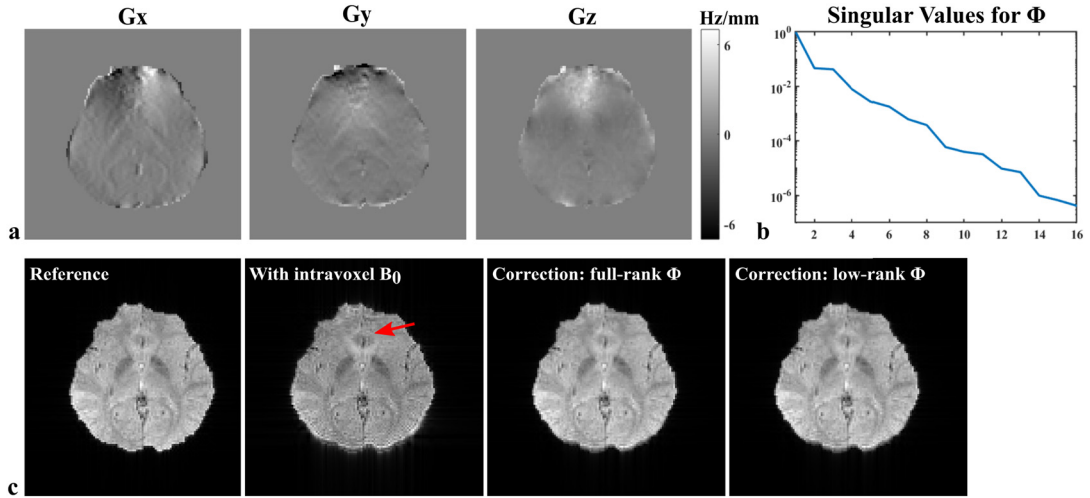


FIGURE 12.13

Illustration of the low-rank property of the encoding operator that incorporates intravoxel B_0 inhomogeneity effects: (a) voxel-field gradients along x , y , and z directions for a particular slice of a practical in vivo B_0 map; (b) the singular values for a Ω_v matrix in log scale (Eqs. (12.15) and (12.27)) for a 2D reconstruction. As can be seen, the singular value decay very rapidly, validating the low-rank assumption. Images in (c) compare images from a direct Fourier reconstruction of the low-resolution data that contains intravoxel B_0 -induced artifacts (second column), corrected reconstruction using the full B_v (third column), and rank-8 approximated Ω_v (last column). The low-rank operator yielded almost the same corrected reconstruction as using the full Ω_v .

with $\mathbf{u}_l$ and a final arbitrary sampling masking matrix $\mathbf{B}$. Such an approximation results in a reduction of computational complexity from $\mathcal{O}(M \times N)$ to $\mathcal{O}(M \times L + N \times L + L \times N \log N)$, compared to evaluating $\mathbf{E}$ directly [19]. Since L is typically orders-of-magnitude smaller than M and N , the reduction is dramatic (Fig. 12.13). It also allows for flexible uses of any regularization functional in the reconstruction formulation for B_0 correction, and is not limited by the PSF as required by the image post-processing based methods.

Fig. 12.13 shows the intra-voxel field gradients from a practical field map (Fig. 12.13a), the low-rank property of B_v constructed from these maps (Fig. 12.13b); and also a set of simulated intravoxel B_0 inhomogeneity corrected reconstructions (Fig. 12.13c). The low-rank approximated operator produced reconstruction with negligible differences from reconstruction using the full Ω_v matrix. However, the low-rank based implementation made the reconstruction 1000 times faster.

12.6 K-space trajectory distortions

The linear portion of the gradients in magnetic-field inhomogeneity, addressed by modeling in the previous section, have an alternative interpretation that helps us to understand their impact. Just like the linear imaging gradients applied to navigate planned sampling trajectories in k-space, the linear

field inhomogeneity terms can be thought of as providing additional k-space encoding. Since the field inhomogeneity gradients are varying across the object, they can result in a unique k-space trajectory for each individual voxel in an image. When the field inhomogeneity gradients are added in, as in Fig. 12.5, the k-space trajectory can deviate significantly from its intended trajectory resulting in sampling errors and timing errors that can have a large impact on images.

Trajectories can be significantly shifted in k-space away from the origin, especially with a long echo time that allows the magnetic field-inhomogeneity gradients, which are constant during the entire acquisition, to accumulate significant encoding even before the planned k-space trajectory starts. If the shift is far enough, then the center of k-space may not even be sampled by the effective trajectory (field-inhomogeneity trajectory plus the planned trajectory). This is an additional interpretation for the signal-loss artifact due to dephasing. It also suggests another approach to address the dephasing artifact—sampling further in k-space than is needed for the imaging requirements. This would also be equivalent to the approach of increasing the spatial resolution (making the voxels smaller) to decrease the dephasing.

An additional issue with the effective trajectory is the timing at which the center of k-space is sampled, which is called the echo time, TE. The TE determines the base contrast for an image and it is based on where the signals, in the absence of magnetic field-inhomogeneity gradients, would refocus from the applied imaging gradients. For a spiral-out acquisition, the TE is at the start of the application of the imaging gradients. However, if the magnetic field inhomogeneity gradients have been allowed to travel outward in k-space for some time, such as in a gradient echo acquisition for functional MRI, then the time at which the effective k-space trajectory hits the center of k-space will be delayed to a later time in the readout. For functional MRI, the signal level is dependent on the echo time, with larger echo times resulting in larger BOLD weighting [69–71]. This can potentially result in differential BOLD signals due to differences in magnetic-field inhomogeneity that could be sensitive to acquisition timing, head orientation, or even anatomical changes across subjects [71].

Before closing this section, we note that sources other than magnetic-field inhomogeneity, such as gradient response functions (e.g. [72,73]) and nonlinearity (e.g. [74]), can also lead to k-space distortion and are important to consider for non-Cartesian trajectories. However, they are beyond the scope of this chapter which focuses on the spatially varying k-space trajectories induced by gradients in the magnetic-field inhomogeneity.

12.7 Measuring the field map

When incorporating the magnetic field-inhomogeneity map into the image reconstruction, there must first be a measurement of the magnetic field map, i.e., the spatial distribution of magnetic field inside the sample that you are imaging. This could be done, as was done earlier in the chapter, by estimating it with flipped PE-axis acquisitions to measure distortion with known sequence parameters. However, this only works for situations with field-inhomogeneity artifacts resulting in simple geometric shifts. More generally, the magnetic field-inhomogeneity map is measured by acquiring a multi-echo acquisition and examining the phase evolution at each voxel to infer its off-resonant frequency [75]. This can be done at the beginning of the scan session for a particular slice prescription because it is fairly stable throughout the acquisition. Depending on the sensitivity of the acquisition to magnetic field-inhomogeneity effects, a dynamic estimation of the field map may be necessary to track field changes during the scan.

In [76], a joint estimation framework through the iterative reconstruction scheme was used, resulting in dynamically estimated magnetic-field maps during a functional MRI scan. These dynamic field maps show the oscillations in the magnetic field in the brain during breathing, due to the movement of the chest wall, with a few Hz at 3 T and stronger at ultrahigh fields (e.g., 7 T) [77].

Sequences to measure the field map should be set up so as to not experience significant image distortions or other artifacts due to magnetic-field inhomogeneity. This would mean multishot, high-bandwidth acquisitions that use short echo times and thin slices. In order for the magnetic field to cause an accumulation of phase, gradient echo acquisitions are needed. But, often this is done with an asymmetric spin echo where a reference image is acquired at the spin echo and the other echos are acquired with slight (1–2 ms) shifts away from the echo. At least two images with different phase accumulation are required, but additional echoes can help in obtaining more accurate field maps. Additionally, the field map-estimation approach could be a simple fit of the linear phase accumulation, or it could incorporate prior constraints on the smoothness of the field maps, such as in [78].

Measuring the field maps repeatedly to account for the dynamic B_0 alterations can be time consuming and may not accurately capture the intra-scan field variations (e.g., due to eddy currents or physiological motions). Recently, specialized field probes have emerged as a promising approach to monitor and estimate these spatiotemporal B_0 field changes in real-time [79–81]. The key assumption for using these field probes are that the spatiotemporal B_0 variations follow a partially separable function representation (with space and time partial separability [82]) and that the spatial components are spherical harmonics [79,83]. Therefore, the temporally varying coefficients can be estimated using limited FIDs from a small number of probes placed around the imaging objects. These field estimates, when incorporated into the image reconstruction process, can offer substantially improved image quality, especially for gradient-echo-based acquisitions at ultrahigh fields [83–85].

12.8 Summary

The standard Fourier imaging for MRI assumes a uniform background magnetic field, which is disrupted in practice by the magnetic properties of the sample that is being imaged. This disruption does not require large ferrous objects to be in the imaging object, and it occurs in every region of the body with large variations of magnetic susceptibility, such as those containing both air and tissues. Uncorrected magnetic-field inhomogeneity can result in image distortions, signal loss, and inadequate k-space sampling due to trajectory deviations. Careful consideration of the physics of magnetic-field inhomogeneity is required in order to minimize its impact through proper choices of imaging sequences and parameters. Several image reconstruction approaches are available that model the effects of magnetic-field inhomogeneity and produce corrected images, as long as the disruptions are not severe. Careful consideration of field inhomogeneity will be of even greater importance for ultrahigh field imaging applications.

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